

Introduction to Operating Systems

An operating system (OS) manages computer hardware and provides services for programs. Its key responsibilities are process management, memory management, file systems and device management.

Process Management

A process is a program in execution. The OS scheduler decides which process runs using algorithms such as First-Come-First-Served, Shortest Job First, Round Robin and priority scheduling. Context switching saves and restores CPU state. Processes communicate through inter-process communication (IPC) mechanisms like pipes, message queues and shared memory. Threads are lightweight units of execution inside a process.

Synchronization and Deadlocks

Concurrent processes need synchronization to avoid race conditions. Critical sections are protected with mutexes, semaphores and monitors. A deadlock occurs when processes wait on each other forever; the four Coffman conditions are mutual exclusion, hold and wait, no preemption and circular wait. The Banker's algorithm is used for deadlock avoidance.

Memory Management

Virtual memory gives each process its own address space. Paging divides memory into fixed-size frames, and the page table maps virtual pages to physical frames. The Translation Lookaside Buffer (TLB) caches translations. Page replacement algorithms include FIFO, LRU and Optimal. Thrashing happens when the system spends more time paging than executing.

File Systems and I/O

File systems organise data into files and directories using structures like inodes and the File Allocation Table. Disk scheduling algorithms such as SCAN and C-SCAN reduce seek time. Device drivers and interrupts let the OS communicate with hardware, and DMA transfers data without CPU involvement.